Practical 3: Data Visualization from ETL Process

Aim

To create **data visualizations (charts and graphs)** in Power BI using the data loaded through the ETL process.

BI

What is Data Visualization?

Data visualization means **showing data using charts, graphs, and maps** so that it is easier to understand.

Example:

Product	Units
Laptop	50

Instead of a table, we create a **bar chart** to understand the data better.

Step-by-Step Practical

Step 1: Open Power BI Project

Open the **Power BI file** where you already loaded the **Products data** in Practical 2.

You will see the **Fields panel on the right side**.

Step 2: Create Units in Stock Chart

1. From **Fields panel**, drag **UnitsInStock** to the blank canvas.
2. A **table visualization** will appear.
3. Now drag **ProductName** to the **Axis** section.

Power BI will create a **bar chart showing Units in Stock by Product**.

BI

Example:

Laptop → 50
Mouse → 120
Keyboard → 80

Step 3: Sort the Chart

1. Click the chart.

2. Click **three dots (⋮)** on the top right.
3. Select **Sort by UnitsInStock**.

Now the chart is sorted by stock values.

Step 4: Create Sales Chart

1. Drag **OrderDate** to the canvas.
2. Drag **LineTotal** to the chart.
3. Select **Line Chart** from visualization options.

This creates a **line chart showing total sales over time**.

BI

Step 5: Create Map Visualization

1. Drag **ShipCountry** to the canvas.
2. Power BI automatically creates a **map**.
3. Drag **LineTotal** to **Values**.

Now each country will show **total sales using circles on the map**.

BI

Step 6: Interact with Visualization

Click any country (for example **Canada**).

Power BI automatically **filters other charts based on that country**.

BI

This helps analyze data easily.

Result

Charts and maps are created using Power BI to analyze product stock and sales data.